

Section: Particles and Waves

Topic: The Standard Model

A simplified version of the Standard Model of particle physics is shown.
It includes three generations of matter particles: **quarks** and **leptons**, and force-carrying particles.

(a) Quarks combine to form hadrons.

State the names of the two types of hadron.

- ✔ The two types of hadrons are:
- **Baryons** (e.g. protons and neutrons) — made of **three quarks**
 - **Mesons** — made of **a quark and an antiquark**

(b) State the names of the two types of lepton in the first generation.

- ✔ The first generation leptons are:
- **Electron**
 - **Electron neutrino**

(c) A pion is a meson.

Give one possible combination of a quark and antiquark that could make up a pion.

- ✔ A valid example:
 $u\bar{d}$ (up and anti – down)

Other acceptable answers include:

- $d\bar{u}$
- $u\bar{u}$
- $d\bar{d}$

(Note: different pions have different combinations.)

🧠 Revision Tips

- **Baryons** = 3 quarks (e.g. uud = proton, udd = neutron)
- **Mesons** = quark + antiquark
- The **first generation** is the one we commonly observe in matter: electron + electron neutrino
- Quark names: up, down, strange, charm, bottom, top
- Quarks carry **fractional charges**; leptons are **fundamental** (not made of smaller particles)